1. Techniques to Avoid Saturated Neurons:

A. weight initialization

B. input standardization.

C. batch normalization.

2. Weight Initialization:

if a single neuron receives a large number of inputs, and its weights are too large,
the value of weighted sum can easily become extremely large or small which causes the neuron to saturate.
therefore, the larger the input number of a single neuron has, the smaller the value of weights of a single neuron should be initially set in the specific layer,
we can achieve this using different types of initializers:
- glorot_uniform(), which is recommended for hyperbolic tangent function or sigmoid function.
- he_normal(), which is recommended for rectified linear unit function.

3. Rectified Linear Unit Function (ReLU()):

A. a type of activation function

B. $f(z) = \max(0, z) = \begin{cases} 0, & z \le 0 \\ z, & z > 0 \end{cases}$

4. Input Standardization:

before the inputs of training dataset / test dataset are fed into the neuron network,
value of the inputs will be standardized (= subtract mean and are divided bi standard deviation) to ensure that value of them is around 0,
reducing the risk of saturated neuron in the input layer.

5. Batch Normalization:

A. internal standardization: value of weighted sum functions or activation functions of all neuron in the hidden layer per batch size will be standardized. (= subtract mean and are divided bi standard deviation)

B. learnable restoration parameters: to address a counterintuitive problem that batch normalization might cancel out the features of original value prior to standardization,
batch normalization includes automatically adjustable parameters (= γ for scaling or β for shifting), this allows the neural network to decide whether to delete the effect of simple normalization,
dynamically finding the most suitable numerical distribution.

C. flexible placement configuration: in implementing neural network architecture, batch normalization offer flexibility in its placement,
it can be placed before activation functions to process value of weighted sum functions,
or after activation functions to process value of activation functions